Sheet Nr. 3:

Exercice 1: Substitution Method

$T(n) = T(\lfloor \sqrt{n} \rfloor) + 1$ with $T(3) = T(2) = T(1) = 0$

claim: $T(n) \in O(\log \log (n))$

So we show for a constant $c > 0$, $T(n) \leq c \cdot \log(\log(n))$

Induction on n:

Induction Hypothesis: $T(k) \leq c \cdot \log(\log(k))$ for all $k < n$

we know $T(n) = T(\lfloor \sqrt{n} \rfloor) + 1 \overset{IH}{\leq} c \cdot \log(\log(\sqrt{n})) + 1$

$$= c \cdot \log(\log(n^{\frac{1}{2}})) + 1$$

$$= c \cdot \left[\log\left(\left(\tfrac{1}{2}\right) \log(n)\right)\right] + 1$$

$$= c \cdot \left(\log\left(\tfrac{1}{2}\right) + \log(\log(n))\right) + 1$$

$$= c \cdot (-1 + \log \log (n)) + 1$$

$$= c \log(\log(n)) - c + 1$$

$$\leq c \cdot \log(\log(n)) \quad \text{if } c \geq 1$$

$\Rightarrow$ claim holds. therefore $T(n) \in O(\log(\log(n)))$

Exercice 2: Mastertheorem

a) $T(n) = 2 \cdot T(\frac{n}{2}) + n \quad , T(1) = 1$

. $a = 2 \geq 1 \quad , b = 2 > 1 \quad , f(n) = n$

. $n^{\log_b a} = n^{\log_2 2} = n^1 = n$

. $f(n) = n$ wich means $f(n) \in \Theta(n^{\log_2 2})$

. this matches case 2.

$\Rightarrow T(n) = \Theta(n^{\log_2 2} \log n) = \Theta(n \log n)$

b) $T(n) = 4 \cdot T(\frac{n}{2}) + \sqrt{n} \quad , T(1) = 1$

. $a = 4 \geq 1 \quad , b = 2 > 1 \quad , f(n) = \sqrt{n}$

. $n^{\log_b a} = n^{\log_2 4} = n^2$

. $f(n) = \sqrt{n} = n^{0.5} \Rightarrow f(n) \in O(n^{2-\varepsilon})$ for $\varepsilon = 1.5 > 0$

$\rightarrow f(n)$ is polynomially smaller: this matches case 1

$\Rightarrow T(n) = \Theta(n^2)$

c) $T(n) = 2 \cdot T(\frac{n}{2}) + f(n)$, where $f(n) = dn^3$ and $d > 0$

. $a = 2 \geq 1, \quad b = 2 > 1 \quad , f(n) = d.n^3$

. $n^{\log_b 2} = n^{\log_2 2} = n.$

$f(n) = dn^3 \Rightarrow f(n) \in \Omega(n^{1+\varepsilon})$ for $\varepsilon = 2 > 0$

$\rightarrow f(n)$ is polynomially larger : this matches case 3

. Regularity conditions must hold: $a . f(\frac{n}{b}) \leq c . f(n)$ for $c < 1$.

$2 . d(\frac{n}{2})^3 = 2.d.\frac{n^3}{8} = \frac{1}{4} dn^3 \overset{!}{\leq} c..dn^3$

This holds true for any c where $\frac{1}{4} \leq c < 1$ ✓

$\Rightarrow$ The regularity condition is satisfied, therefore

$T(n) = \Theta(dn^3) = \Theta(n^3)$

d) $T(n) = 2 \cdot T\left(\frac{n}{2}\right) + \frac{n}{\log_2 n}$

- $a = 2 \geq 1$, $b = 2 > 1$. $f(n) = \frac{n}{\log_2 n}$
- $n^{\log_b a} = n^{\log_2 2} = n$
- $f(n) = \frac{n}{\log_2 n}$ is not polynomialy smaller than n ($f \notin O(n^{1-\varepsilon})$ for $\varepsilon > 0$)

$\rightarrow$ not case 1. and not case 2 ($f \notin \Theta(n)$)

- $f(n) = \frac{n}{\log_2 n}$ ist not polynomially larger than n ($f \notin \Omega(n^{1+\varepsilon})$ for $\varepsilon > 0$)

$\rightarrow$ not case 3.

$\Rightarrow$ The Master Theorem cannot be applied.

Recursion Tree Method.

level	# nodes	Level contribution
0	$2^0 = 1$	$\frac{n}{\log_2(n)}$
1	$2^1 = 2$	$2 \cdot \frac{n/2}{\log_2(\frac{n}{2})} = \frac{n}{\log_2(n) - \log_2(2)} = \frac{n}{\log_2(n) - 1}$
2	$2^2 = 4$	$4 \cdot \frac{n/4}{\log_2(\frac{n}{4})} = \frac{n}{\log_2(n) - \log_2(4)} = \frac{n}{\log_2(n) - 2}$
$\vdots$	$\vdots$	$\vdots$
i	2^i	$2^i \cdot \frac{n/2^i}{\log_2(\frac{n}{2^i})} = \frac{n}{\log_2(n) - \log_2(2^i)} = \frac{n}{\log_2(n) - i}$
$\vdots$		
$\log_2(n)$	$2^{\log_2(n)} = n$	n $(T(1) = 1)$

$$\rightarrow T(n) = \overset{\text{lowest level}}{n} + \underbrace{\sum_{i=0}^{\log_2(n)-1} \frac{n}{\log_2(n) - i}}_{\text{other levels}} = n + n \sum_{i=0}^{\log_2(n)-1} \frac{1}{\log_2(n) - i}$$

let $k = \log_2(n) - i$ $\Rightarrow$ new sum bounds:
- $k = \log_2(n) - 0 = \log_2(n)$
- $k = \log_2(n) - (\log_2(n) - 1) = 1$

$$\rightarrow T(n) = n + n \sum_{k=1}^{\log_2(n)} \frac{1}{k} \qquad \left(\sum_{k=1}^{x} \frac{1}{k} = \Theta(\log(x)) \text{ : harmonic series}\right)$$

$$= \Theta(n) + \Theta(n \log(\log(n))) = \Theta(n \log(\log(n)))$$

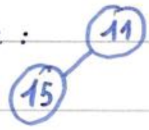
Exercice 3 : Min-heap

1. insert (11)

Tree : 

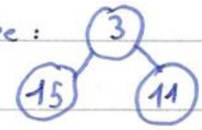
array : [11]

2. insert (15)

Tree : 

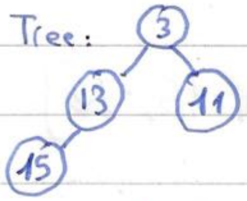
array: [11, 15]

3. insert (3)

Tree : 

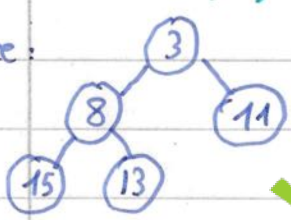
array : [3, 15, 11]

4. insert (13)

Tree : 

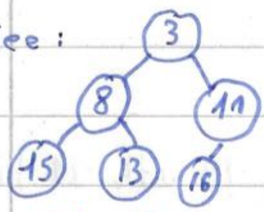
array : [3, 13, 11, 15]

5. insert (8)

Tree : 

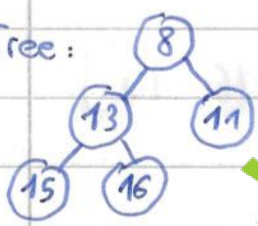
array : [3, 8, 11, 15, 13]

6. insert (16)

Tree : 

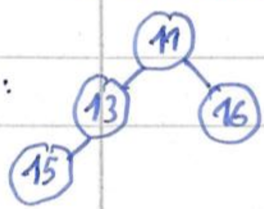
array : [3, 8, 11, 15, 13, 16]

7. Extract min() : remove root, 16 → root, swap 16 with 8 then 13.

Tree : 

array : [8, 13, 11, 15, 16]

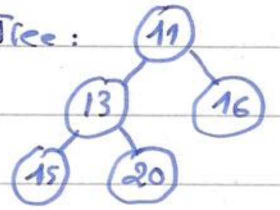
8. Extract min() : remove 8, 16 → root, swap with 11.

Tree : 

array : [11, 13, 16, 15]

9. Insert (20)

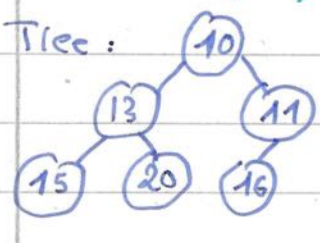
Tree :



array: [11, 13, 16, 15, 20]

10. Insert (10)

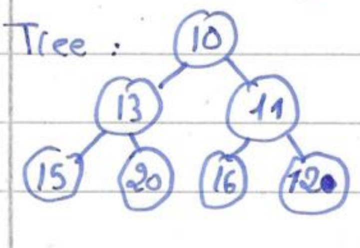
Tree :



array: [10, 13, 11, 15, 20, 16]

11. Insert (12)

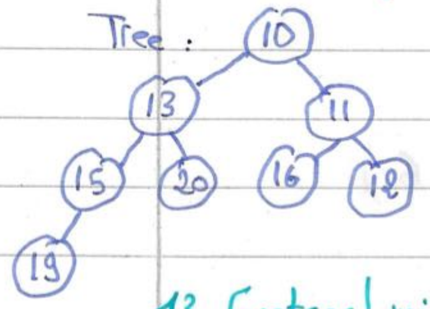
Tree :



array: [10, 13, 11, 15, 20, 16, 12]

12. Insert (19)

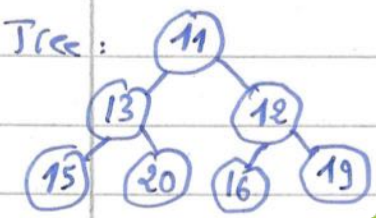
Tree :



array: [10, 13, 11, 15, 20, 16, 12, 19]

13. Extract min () : remove 10, 19 → root, swap 19 with 11 then with 12.

Tree :



array: [11, 13, 12, 15, 20, 16, 19]

Index of comments

1.1 and n >= 2

1.2 (IB) is missing (-0.5)

1.3 and k >= 2

3.1 Explanations are a little vague, would have been better with limes